SU(3) in four dimensions: every constant of this programme, evaluated

With the goal narrowed to $SU(3)$, each bound proved in these notes becomes a number. The group data are

$$N = 3, \quad \dim G = 8, \quad C_2(R_{\rm f}) = \frac{N^2-1}{2N} = \frac{4}{3}, \quad C_2(\text{adj}) = N = 3, \quad Z(G) = \mathbb{Z}_3,$$

in the normalization $\operatorname{tr} T^a T^b = \frac{1}{2}\delta^{ab}$, and the one- and two-loop coefficients are

$$b_0 = \frac{11N}{48\pi^2} = \frac{11}{16\pi^2} = 0.06966, \qquad \frac{b_1}{2b_0^2} = \frac{51}{121} = 0.4215.$$

The consequences, collected: the strong-coupling criterion of the T2 note reads $54/g^4 \le \beta_*$; the gap it gives is $\Delta_{a,L} \ge \frac{2}{3}\gamma g^2\,\hbar c/a$; the unperturbed flux-loop gap is $\frac{8}{3}g^2\hbar c/a$; the Lieb–Robinson velocity is $v \le 96e\,c/g^2 \simeq 261\,c/g^2$; the exact identities are $\Delta V = \frac{16}{3}(3|\mathcal{P}|-V)$ and $|\nabla V|^2 \le 48V$; the number of renormalization doublings from a bare coupling $g^2_{\rm UV} = 1/2$ to a strong-coupling threshold is

$$n \simeq \frac{1}{2b_0 \log 2}\Big(\frac{1}{g^2_{\rm UV}} - \frac{1}{g^2_{\rm thr}}\Big) \le \frac{2}{0.0966} \simeq 21,$$

fewer than the thirty of $SU(2)$ because the coupling runs faster; and the flowed upper bound $4.26\,\hbar c/\sqrt{8t}$ is unchanged, because in the free limit the eight gluon species multiply numerator and denominator equally. Two features of $SU(3)$ that $SU(2)$ lacks are recorded in Section 4: the defining representation is complex, so $\operatorname{tr} U_p$ is complex and the plaquette term keeps only its real part, and the centre is $\mathbb{Z}_3$, so a periodic box carries 27 electric-flux sectors instead of 8. Constants explicit; nothing promoted.

1. Group data

quantity	value	where used
N	3	everywhere
$\dim G$	8	free-limit species count
$C_2(R_{\rm f})$	4/3	flux-loop energy, T2 gap
$C_2(\text{adj})$	3	Jacobian curvature factor
centre	$\mathbb{Z}_3$	flux sectors, valley minima
$\operatorname{tr}(1)$ in $R_{\rm f}$	3	plaquette norm $\|w_p\| \le 4N\hbar c/(ag^2) = 12\hbar c/(ag^2)$
b_0	$11/(16\pi^2) = 0.06966$	running, step count
$b_1/(2b_0^2)$	$51/121 = 0.4215$	the power in $a\Lambda_{\rm lat}$

The lattice Λ parameter of the obligations map §5 is therefore

$$a\Lambda_{\rm lat}(g) = (b_0 g^2)^{-51/121}\, e^{-1/(2b_0 g^2)}\,[1 + O(g^2)], \qquad \frac{1}{2b_0} = \frac{8\pi^2}{11} = 7.180.$$

2. The proved bounds, evaluated

T1. Unchanged in form: unique positive physical ground state and $\delta(g; N_s, SU(3)) > 0$ at every finite lattice and coupling.

T2 (T2 note). The perturbation parameter is

$$\beta = \frac{48N^2}{g^4(N^2-1)} = \frac{48\cdot 9}{8\,g^4} = \frac{54}{g^4},$$

so the criterion $\beta \le \beta_*(3, \{0,1\}^3)$ becomes

$$g \;\ge\; g_0 = \Big(\frac{54}{\beta_*}\Big)^{1/4},$$

and the gap is

$$\Delta_{a,L} \;\ge\; \gamma\,\frac{g^2}{2}\,\frac{4}{3}\,\frac{\hbar c}{a} = \frac{2\gamma}{3}\,g^2\,\frac{\hbar c}{a}.$$

The unperturbed physical excitation, a single plaquette flux loop in the fundamental, costs $2C_2 g^2 \hbar c/a = \frac{8}{3} g^2 \hbar c/a$.

Feynman–Bijl, strong coupling (upper-bound note Corollary 3). At $g = \infty$ the Haar variance is $\mathrm{Var}(\mathrm{Re}\,\mathrm{tr}\,U_p) = \frac{1}{2}$ for $N \ge 3$, so the two-sided statement reads

$$\frac{2\gamma}{3} g^2\,\frac{\hbar c}{a} \;\le\; \Delta_{a,L}^{\mathrm{phys}} \;\le\; 12\,g^2\,\frac{\hbar c}{a}$$

for g beyond both thresholds, a ratio of $18/\gamma$ between the two sides.

Lieb–Robinson (note). Four plaquettes per link and $\|w_p\| \le 12\hbar c/(ag^2)$ give $J \le 48\hbar c/(ag^2)$ and

$$v = \frac{2eaJ}{\hbar} \;\le\; \frac{96\,e\,c}{g^2} \;\simeq\; \frac{261\,c}{g^2},$$

so the lattice light cone exceeds c for $g^2 < 261$, that is throughout any regime of interest, and the strong-coupling correlation length is $\xi \le C'\,a\,/(\gamma g^4)\cdot\frac{9}{4}$.

Exact identities (note). With $C_2 = 4/3$ and $N = 3$,

$$\Delta V = \frac{16}{3}(3|\mathcal{P}| - V), \qquad |\nabla V|^2 \le 16NV = 48\,V,$$

and the ground-state sum rule of that note holds with $2C_2 = 8/3$.

Blocking (criterion note). The block criterion is $\beta_{\mathrm{block}} = 72M^2/(g^2\delta(g;M))$ and the direct one is $\beta_{\mathrm{direct}} = 24/g^4$, so the ratio at strong coupling is $3M^2/8$ as before, blocking losing by $3/2$ already at $M = 2$.

Small volume (G07 Proposition 7). The constant-mode sector has $D = 3$ and gauge group $SU(3)$, so the matrix model runs over $\vec{x}_i \in \mathbb{R}^8$ with the structure constants of $SU(3)$ in place of the cross product; the gap keeps the form $\delta_1^{(3)} g^{2/3}\hbar c/L$ with a pure number $\delta_1^{(3)}$ specific to the group.

3. The step count

With $2b_0 \log 2 = 2(0.06966)(0.6931) = 0.09657$, the number of renormalization doublings needed to run from a bare coupling to a strong-coupling threshold is

$$n = \frac{1}{0.09657}\Big(\frac{1}{g_{\mathrm{UV}}^2} - \frac{1}{g_{\mathrm{thr}}^2}\Big).$$

g_{UV}^2	n to $1/g_{\mathrm{thr}}^2 \to 0$	scale ratio 2^n
1	10.4	1.4×10^3
1/2	20.7	1.7×10^6
1/4	41.4	2.9×10^{12}

For $SU(2)$ the same table reads 15.5, 31.1, 62.1, since b_0 is smaller by $2/3$. **The $SU(3)$ problem needs about a third fewer steps than** $SU(2)$ at the same bare coupling, which is the only quantitative respect in which the physical case is easier.

4. Two features specific to $SU(3)$

The defining representation is complex. For $SU(2)$, $\operatorname{tr} U_p$ is real, and the plaquette term is the full trace; for $SU(3)$ it is complex and the Wilson term keeps $\operatorname{Re}\operatorname{tr} U_p$. Consequences: the bound $\int(\operatorname{tr} U)^2 dU = 0$ used in the upper-bound note Corollary 3 holds for $N \geq 3$ and gives Haar variance $\frac{1}{2}$ rather than 1; and charge conjugation is a nontrivial symmetry, so the spectrum splits into C-even and C-odd sectors, and the glueball channel relevant to the gap is C-even.

The centre is $\mathbb{Z}_3$. The valley minima of the valley note §4 lie at holonomies that are central, so there are three per direction and 27 electric-flux sectors on a three-torus, against 8 for $SU(2)$. The tunnelling barrier between them is the same $O(\hbar c/L)$, and the sector structure is what the small-volume spectrum organizes itself by.

5. What does not change

The structural results are group-independent and stand as written:

- the seven closed routes of the position note §3, each closed by an argument that uses no property of the group beyond compactness and non-abelianness;
- the moment hierarchy and the reduction of $m < \infty$ to one flowed correlator, where the free-limit value $4.26\,\hbar c/\sqrt{8t}$ is unchanged because the $\dim G = 8$ species multiply the susceptibility and the gradient term equally;
- the equivalence of T2′ with the absence of a zero-temperature phase transition;
- the three places where the non-abelian structure is isolated, all of which are statements about the commutator and hold for $SU(3)$ with the $SU(3)$ structure constants.

6. Consequence for STATE

Specializing to $SU(3)$ fixes every constant and shortens the step count to about twenty-one doublings from $g^2_{\mathrm{UV}} = 1/2$. It changes no structural conclusion, and the open obligations are those of the position note §5. The next work should be carried out with these numbers in place, and any new bound should be reported in them.